

Question

As shown in Figure 1, the reactant undergoes the following reaction mechanism to obtain intermediate **A**⁺, intermediate **B**, and final product **C**, respectively:



Fig. 1, the figure shows a series of consecutive reaction mechanisms. The reactant is described in SMILES as: CS(/C(SC)=C/C(F)(F)F)=O. The reactant reacts with Tf_2O , losing one molecule of TfO^- , and then reacts with phenol, losing one molecule of TfOH , to obtain intermediate **A**⁺. Intermediate **A**⁺ undergoes rearrangement and deprotonation to obtain intermediate **B**. Intermediate **B** undergoes elimination to obtain final product **C**.

Speculate on the reaction mechanism and the structures of intermediate **A**⁺, **B**, and final product **C**.

The following statements are made:

- A**⁺ contains two carbon-oxygen bonds.
- B** contains one sulfur-oxygen bond.
- The product of the number of atoms within each six-membered ring or smaller in **C**, multiplied by the number of carbon-non-carbon non-hydrogen chemical bonds (multiple bonds counted as one bond), is 210.
- The oxidation state of the sulfur atom decreases during the entire reaction process.

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **K**

Detailed Explanation

In the first step, triflic anhydride can activate the oxygen atom of the reactant, yielding the activated sulfinyl group. It can undergo a nucleophilic substitution reaction with phenol, where phenol attacks the sulfur atom, substituting triflate, to obtain intermediate $\mathbf{A}^+$, the structure of which is shown in Figure 2:

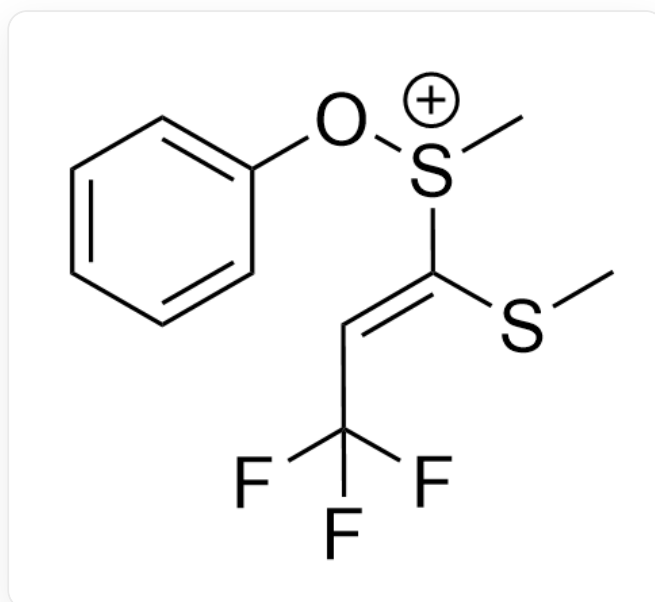


Fig. 2, the molecule in the figure is described by SMILES as: C[S+](OC1=CC=CC=C1)/C(SC)=C/C(F)(F)F

CHECKPOINT

1 PTS

Oxygen atom is activated by triflic anhydride, phenol substitutes TfOH, yielding intermediate $\mathbf{A}^+$ described by SMILES as: C[S+](OC1=CC=CC=C1)/C(SC)=C/C(F)(F)F

$\mathbf{A}^+$ contains only one carbon-oxygen bond, statement 1 is incorrect.

According to the question, intermediate $\mathbf{A}^+$ undergoes a rearrangement reaction in the next step. Observing the molecular structure, the sulfur atom is electrophilic, but there is no good nucleophilic site within the molecule. A more likely rearrangement is that the sulfur-oxygen bond undergoes a concerted [3, 3]-sigma migration rearrangement reaction through the double bond and the benzene ring, forming a carbon-carbon bond, yielding the intermediate in Figure 3:

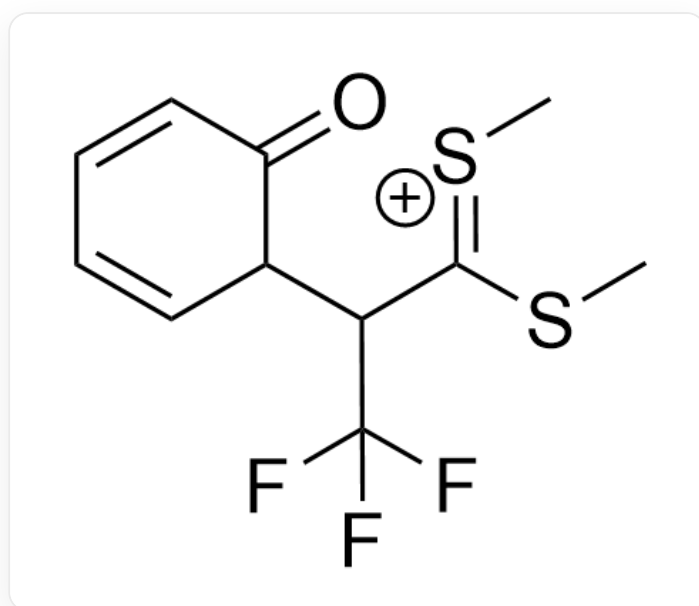


Fig. 3, the molecule in the figure is described by SMILES as: C/[S+]=C(C(C1C(C=CC=C1)=O)C(F)(F)F)\SC

CHECKPOINT

1 PTS

Intermediate $\mathbf{A}^+$ undergoes a concerted [3, 3]-sigma migration rearrangement reaction, yielding the intermediate in Figure 3 described by SMILES as: C/[S+]=C(C(C1C(C=CC=C1)=O)C(F)(F)F)\SC

This intermediate quickly regains aromaticity to form a hydroxyl group, the hydroxyl group undergoes nucleophilic addition to the carbon-sulfur double bond and deprotonates to obtain intermediate $\mathbf{B}$, the structure of which is shown in Figure 4:

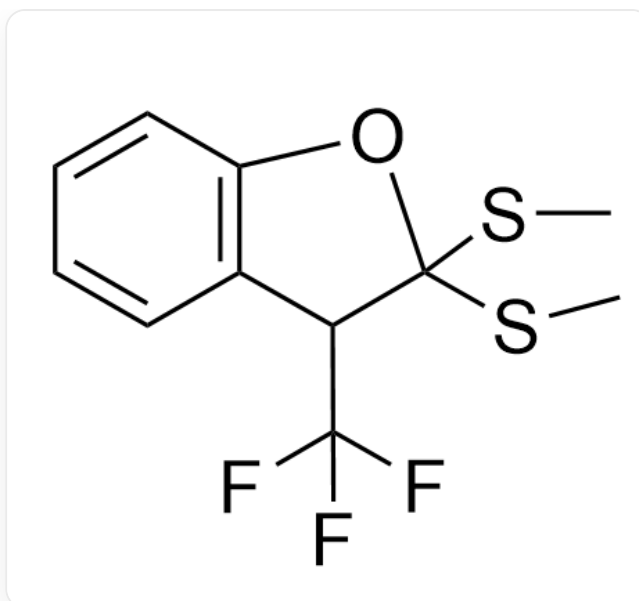


Fig. 4, the molecule in the figure is described by SMILES as: CSC1(OC(C=CC=C2)=C2C1C(F)(F)F)SC

CHECKPOINT

1 PTS

The intermediate in Figure 3 quickly regains aromaticity to form a hydroxyl group, the hydroxyl group undergoes nucleophilic addition to the carbon-sulfur double bond and deprotonates to obtain intermediate **B**, described by SMILES as: CSC1(OC(C=CC=C2)=C2C1C(F)(F)F)SC

B does not contain a sulfur-oxygen bond, statement 2 is incorrect.

Intermediate **B** eliminates a molecule of methanethiol, yielding the aromatic benzofuran structure **C**, as shown in Figure 5:

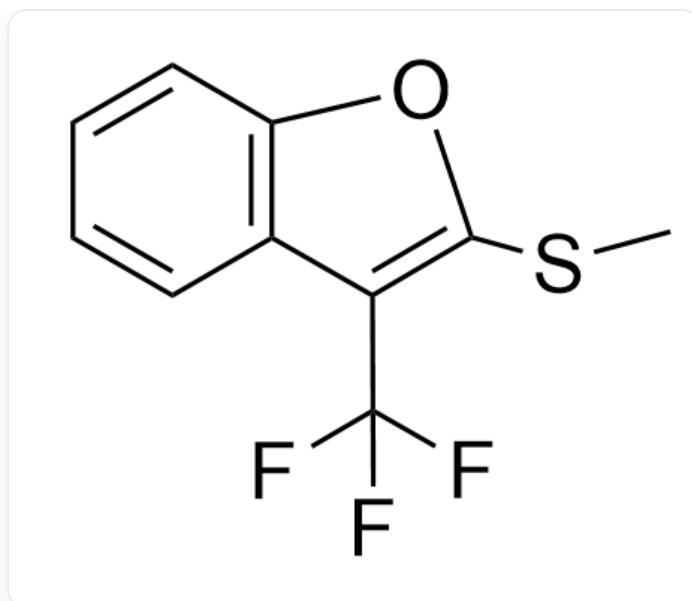


Fig. 5, the molecule in the figure is described by SMILES as: FC(F)(F)C1=C(OC2=C1C=CC=C2)SC

CHECKPOINT

1 PTS

Intermediate **B** eliminates a molecule of methanethiol, yielding **C** described by SMILES as: FC(F)(F)C1=C(OC2=C1C=CC=C2)SC

In **C**, the number of atoms within the six-membered and five-membered rings are 6 and 5, respectively, and the number of carbon-non-carbon non-hydrogen chemical bonds (multiple bonds counted as one bond) is 7. The product of these three is 210, statement 3 is correct.

The sulfur atom of sulfoxide in the reactant forms two carbon-sulfur bonds and one oxygen-sulfur bond, and the final product breaks the oxygen-sulfur bond, leaving only two carbon-sulfur bonds. The entire reaction process is equivalent to the sulfur atom being reduced, and the oxidation state is reduced from +4 to +2, statement 4 is correct.